DBMS – SQL QUERY IMPLEMENTATION FOR E-COMMERCE DATABASE

Project Title : Lenskart E-Commerce Order Management System

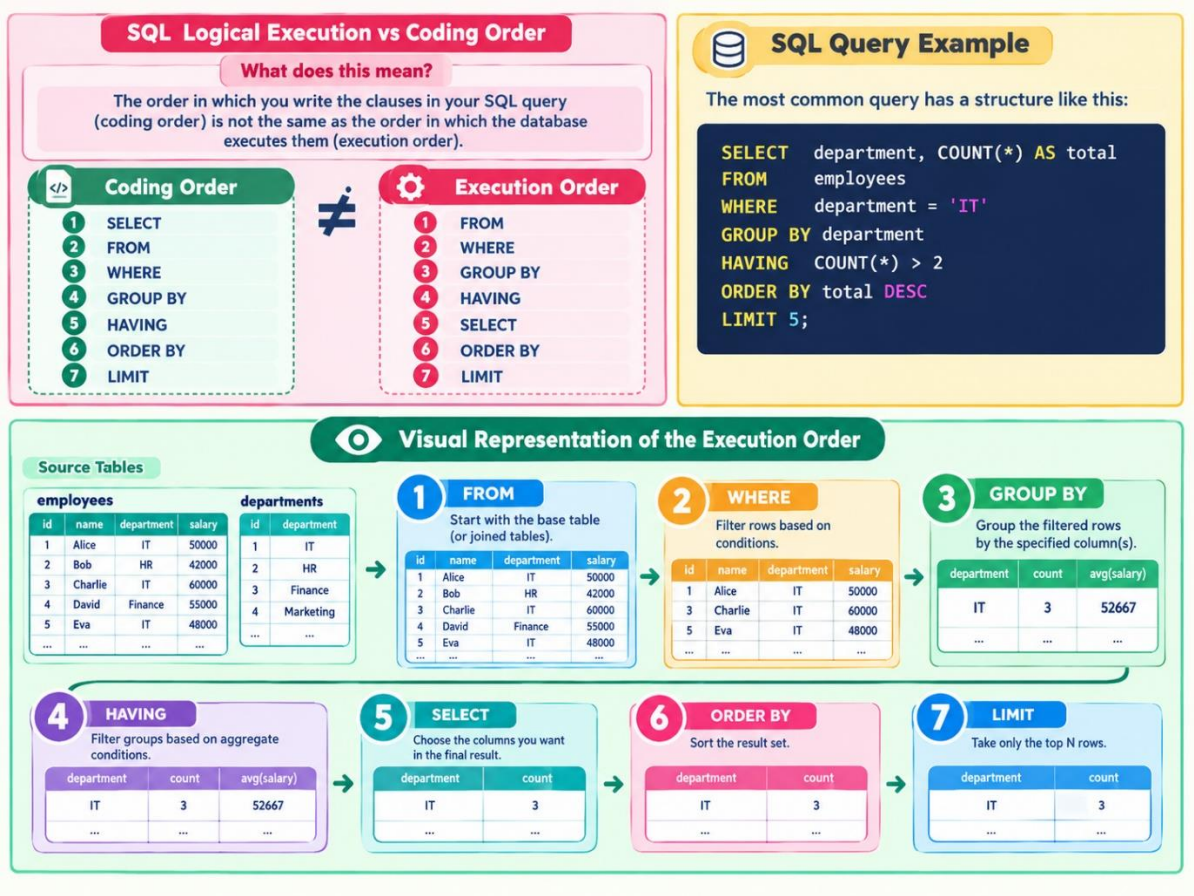
Objectives

1. Perform SELECT, WHERE, ORDER BY, and DISTINCT queries.
2. Search products based on price,category, and availability.
3. Retrieve customer and product information.
4. Apply filtering conditions.
5. Generate basic business reports.

1. Basic SQL Queries

1.1 SELECT QUERY

The SELECT statement is used to retrieve data from a table.



Display all products

```
SQL> select * from product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID FRAME_TYPE
----------- ---------- --------------------------------------------------
     PRICE      STOCK POWER_COMP
---------- ---------- ----------
       201
Classic Eyeglasses
         1        101 Full Frame
      1200         50 Yes


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID FRAME_TYPE
----------- ---------- --------------------------------------------------
     PRICE      STOCK POWER_COMP
---------- ---------- ----------
       202
Stylish Sunglasses
         2        102 Rimless
      1500         30 No


PRODUCT_ID
----------
```

```
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID FRAME_TYPE
----------- ---------- --------------------------------------------------
     PRICE      STOCK POWER_COMP
---------- ---------- ----------
       203
Reading Glasses
         1        103 Half Frame
       800         45 Yes


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID FRAME_TYPE
----------- ---------- --------------------------------------------------
     PRICE      STOCK POWER_COMP
---------- ---------- ----------
       204
Contact Lenses
         3        104 None
      1000         25 Yes


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID FRAME_TYPE
----------- ---------- --------------------------------------------------
```

Display selected product data

```
SQL> select product_id, product_name, price from product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       201
Classic Eyeglasses
      1200

       202
Stylish Sunglasses
      1500

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       203
Reading Glasses
       800

       204
Contact Lenses

PRODUCT_ID
----------
PRODUCT_NAME
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1000

       205
Blue Light Glasses
      1300
```

1.2 WHERE Clause

The WHERE clause is used to filter records based on a condition.

Find products with price less than 1000

```
SQL> select * from product where price < 1000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID FRAME_TYPE
----------- ---------- --------------------------------------------------
     PRICE      STOCK POWER_COMP
---------- ---------- ----------
       203
Reading Glasses
         1        103 Half Frame
       800         45 Yes
```

Find products with price greater than 1000

```
SQL> select * from product where price > 1000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID FRAME_TYPE
----------- ---------- --------------------------------------------------
     PRICE      STOCK POWER_COMP
---------- ---------- ----------
       201
Classic Eyeglasses
         1        101 Full Frame
      1200         50 Yes


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID FRAME_TYPE
----------- ---------- --------------------------------------------------
     PRICE      STOCK POWER_COMP
---------- ---------- ----------
       202
Stylish Sunglasses
         2        102 Rimless
      1500         30 No


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID FRAME_TYPE
----------- ---------- --------------------------------------------------
     PRICE      STOCK POWER_COMP
---------- ---------- ----------
       205
Blue Light Glasses
         1        105 Full Frame
      1300         60 Yes
```

Find products with price greater than 100

```
SQL> select * from product where price > 100;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID FRAME_TYPE
----------- ---------- --------------------------------------------------
     PRICE      STOCK POWER_COMP
---------- ---------- ----------
       201
Classic Eyeglasses
         1        101 Full Frame
      1200         50 Yes


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID FRAME_TYPE
----------- ---------- --------------------------------------------------
     PRICE      STOCK POWER_COMP
---------- ---------- ----------
       202
Stylish Sunglasses
          2        102 Rimless
      1500         30 No


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID FRAME_TYPE
----------- ---------- --------------------------------------------------
     PRICE      STOCK POWER_COMP
---------- ---------- ----------
       203
```

1.3 ORDER BY

The ORDER BY clause is used to sort the query results.

Products from lowest to highest price

```
SQL> SELECT product_name, product_id, price
  2  FROM product
  3  ORDER BY price ASC;

PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT_ID      PRICE
---------- ----------
Reading Glasses
       203        800

Contact Lenses
       204       1000

Classic Eyeglasses
       201       1200


PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT_ID      PRICE
---------- ----------
Blue Light Glasses
       205       1300

Stylish Sunglasses
       202       1500
```

Products from highest to lowest price

```
SQL> SELECT product_name, product_id, price
  2  FROM product
  3  ORDER BY price DESC;

PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT_ID      PRICE
---------- ----------
Stylish Sunglasses
       202       1500

Blue Light Glasses
       205       1300

Classic Eyeglasses
       201       1200


PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT_ID      PRICE
---------- ----------
Contact Lenses
       204       1000

Reading Glasses
       203        800
```

1.4 DISTINCT

The DISTINCT keyword is used to display unique values.

Display unique categories

```
SQL> select distinct(price) from product;

     PRICE
----------
      1200
      1500
       800
      1000
      1300
```

2. Search Products

2.1 Search Products Based on Price

```
SQL> select * from product where price between 800 and 1000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID FRAME_TYPE
----------- ---------- --------------------------------------------------
     PRICE      STOCK POWER_COMP
---------- ---------- ----------
       203
Reading Glasses
         1        103 Half Frame
       800         45 Yes


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID FRAME_TYPE
----------- ---------- --------------------------------------------------
     PRICE      STOCK POWER_COMP
---------- ---------- ----------
       204
Contact Lenses
         3        104 None
      1000         25 Yes
```

This displays products whose price is between ₹100 and ₹400.

2.2 Search Products Based on Category

```
SQL> select * from product where category_id = 1;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID FRAME_TYPE
----------- ---------- --------------------------------------------------
     PRICE      STOCK POWER_COMP
---------- ---------- ----------
       201
Classic Eyeglasses
         1        101 Full Frame
      1200         50 Yes


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID FRAME_TYPE
----------- ---------- --------------------------------------------------
     PRICE      STOCK POWER_COMP
---------- ---------- ----------
       203
Reading Glasses
          1        103 Half Frame
       800         45 Yes


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID FRAME_TYPE
----------- ---------- --------------------------------------------------
     PRICE      STOCK POWER_COMP
---------- ---------- ----------
       205
Blue Light Glasses
```

This displays products belonging to category 1.

3. Retrieve Customer Information

Display all customers

```
SQL> select * from customer;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE_NO        DATE_OF_B
--------------- ---------
ADDRESS
--------------------------------------------------------------------------------
          1 Arun Kumar
arun@gmail.com
9876543210      12-AUG-04
Chennai


CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE_NO        DATE_OF_B
--------------- ---------
ADDRESS
--------------------------------------------------------------------------------
          2 Priya
priya@gmail.com
9876543211      20-FEB-05
Madurai


CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE_NO        DATE_OF_B
--------------- ---------
ADDRESS
--------------------------------------------------------------------------------
```

Search for a particular customer

```
SQL> select * from customer where customer_id = 1;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE_NO        DATE_OF_B
--------------- ---------
ADDRESS
--------------------------------------------------------------------------------
          1 Arun Kumar
arun@gmail.com
9876543210      12-AUG-04
Chennai
```

4. Apply Filtering Conditions

Multiple conditions can be combined using AND,OR, and NOT.

4.1 AND Condition

```
SQL> SELECT product_name, product_id, price
  2  FROM product
  3  WHERE price > 1000
  4  AND category_id = 2;

PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT_ID      PRICE
---------- ----------
Stylish Sunglasses
       202       1500
```

This displays products that satisfy both conditions.

4.2 OR Condition

```
SQL> SELECT product_name, product_id, price
  2  FROM product
  3  WHERE price > 1000
  4  OR category_id = 2;

PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT_ID      PRICE
---------- ----------
Classic Eyeglasses
       201       1200

Stylish Sunglasses
       202       1500

Blue Light Glasses
       205       1300
```

This displays products belonging to either category 101 or 102.

4.3 IN Operator

```
SQL> SELECT product_name, product_id, price
  2  FROM product
  3  WHERE category_id IN (1, 2, 3);

PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT_ID      PRICE
---------- ----------
Classic Eyeglasses
       201       1200

Stylish Sunglasses
       202       1500

Reading Glasses
       203        800


PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT_ID      PRICE
---------- ----------
Contact Lenses
       204       1000

Blue Light Glasses
       205       1300
```

4.4 BETWEEN Operator

```
SQL> SELECT product_name, product_id, price
  2  FROM product
  3  WHERE price BETWEEN 1000 AND 5000;

PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT_ID      PRICE
---------- ----------
Classic Eyeglasses
       201       1200

Stylish Sunglasses
       202       1500

Contact Lenses
       204       1000


PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT_ID      PRICE
---------- ----------
Blue Light Glasses
       205       1300
```

5. Basic Business Reports

5.1 Product Price Report

```
SQL> SELECT product_name, product_id, price
  2  FROM product
  3  ORDER BY price ASC;

PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT_ID      PRICE
---------- ----------
Reading Glasses
       203        800

Contact Lenses
       204       1000

Classic Eyeglasses
       201       1200


PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT_ID      PRICE
---------- ----------
Blue Light Glasses
       205       1300

Stylish Sunglasses
       202       1500
```

5.2 Customer Order Report

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.Customer_Name,
  4      o.Order_ID,
  5      o.Order_Date,
  6      o.Total_Amount
  7  FROM Customer c
  8  JOIN Orders o
  9      ON c.Customer_ID = o.Customer_ID
 10  ORDER BY o.Order_Date DESC;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
         10 Harini                                                110
29-SEP-26         3999

          9 Sanjay                                                109
28-SEP-26         2699

          8 Nandhini                                              108
27-SEP-26         1799


CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
          7 Rahul                                                 107
26-SEP-26         2199

          6 Keerthana                                             106
25-SEP-26         1299

          5 Vignesh                                               105
```

5.3 Product Order Report

```
SELECT
    oi.Order_ID,
    oi.Product_ID,
    p.Product_Name,
    oi.Quantity,
    oi.Price AS Unit_Price,
    (oi.Quantity * oi.Price) AS Subtotal
FROM Order_Item oi
JOIN Product p
    ON oi.Product_ID = p.Product_ID
ORDER BY Subtotal DESC;

 ORDER_ID PRODUCT_ID
---------- ----------
PRODUCT_NAME
--------------------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ----------
       110        205
Blue Light Glasses
         2       1300       2600


       101        201
Classic Eyeglasses
         2       1200       2400
```

5.4 Customer-wise Total Orders

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.Customer_Name,
  4      COUNT(o.Order_ID) AS Total_Orders
  5  FROM Customer c
  6  LEFT JOIN Orders o
  7      ON c.Customer_ID = o.Customer_ID
  8  GROUP BY
  9      c.Customer_ID,
 10      c.Customer_Name
 11  ORDER BY Total_Orders DESC;

CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_ORDERS
----------- -------------------------------------------------- ------------
          1 Arun Kumar                                                    1
          2 Priya                                                         1
          3 Karthik                                                       1
          4 Divya                                                         1
         10 Harini                                                        1
          6 Keerthana                                                     1
          7 Rahul                                                         1
          8 Nandhini                                                      1
          9 Sanjay                                                        1
          5 Vignesh                                                       1

10 rows selected.
```

5.5 Customer-wise Total Purchase Amount

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.Customer_Name,
  4      SUM(o.Total_Amount) AS Total_Purchase_Amount
  5  FROM Customer c
  6  JOIN Orders o
  7      ON c.Customer_ID = o.Customer_ID
  8  GROUP BY
  9      c.Customer_ID,
 10      c.Customer_Name
 11  ORDER BY Total_Purchase_Amount DESC;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
TOTAL_PURCHASE_AMOUNT
---------------------
         10 Harini
                3999

          5 Vignesh
                3499

          3 Karthik
                2999


CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
TOTAL_PURCHASE_AMOUNT
---------------------
          9 Sanjay
                2699

          1 Arun Kumar
                2499
```